5 Solve the simultaneous equations

$$3x - 2y = -5$$
$$2x - 4y = 2$$

Double the first equation:

$$6x - 4y = -10$$
$$2x - 4y = 2$$

Subtract: $4x = -12$, so $x = -3$.

1 mark
Eliminate y

Substitute into $2x - 4y = 2$:

$$-6 - 4y = 2$$
$$+6 \qquad +6$$
$$-4y = 8$$
$$\div(-4) \qquad \div(-4)$$
$$y = -2$$

1 mark
Substitute to find y

1 mark
x=−3 and y=−2

Check: $-9 + 4 = -5$; $-6 + 8 = 2$.

$x =$ .....

$y =$ .....

(Total for Question 5 is 3 marks)